AI discovery: workflows, evidence and sensitivities

Modalities use different units. Workload sensitivities and historical clinical benchmarks are shown separately.

STAGE	BEFORE GEN-AI	AI MECHANISMS	WHAT CHANGES
Virtual search / design	Large virtual libraries	Quality + selection	Large libraries predate generative AI. The investment mechanism is better selection and learning.
Physical samples	Program-specific	Modality-dependent	Small molecules can need fewer syntheses per program. Protein libraries and individual profiling are different units.
Assay workload	**1× baseline**	**1× / 2× / 5× / 10×**	**Analyst workload sensitivities, not predicted fully profiled candidate counts. Include an efficiency/no-growth case.**
Clinical development	Phase I programs	Funded clinical programs	Clinical biology, patients, capital and time remain constraints; discovery gains do not establish clinical gains.
Phase I to approval	7.9% / 13.8% (study estimates)	AI advantage unproven	BIO 2011–2020 and Wong 2000–2015 are different cohorts and methods, not an AI comparison.

Tool demand depends on funded programs, assay depth, net inputs and prices.

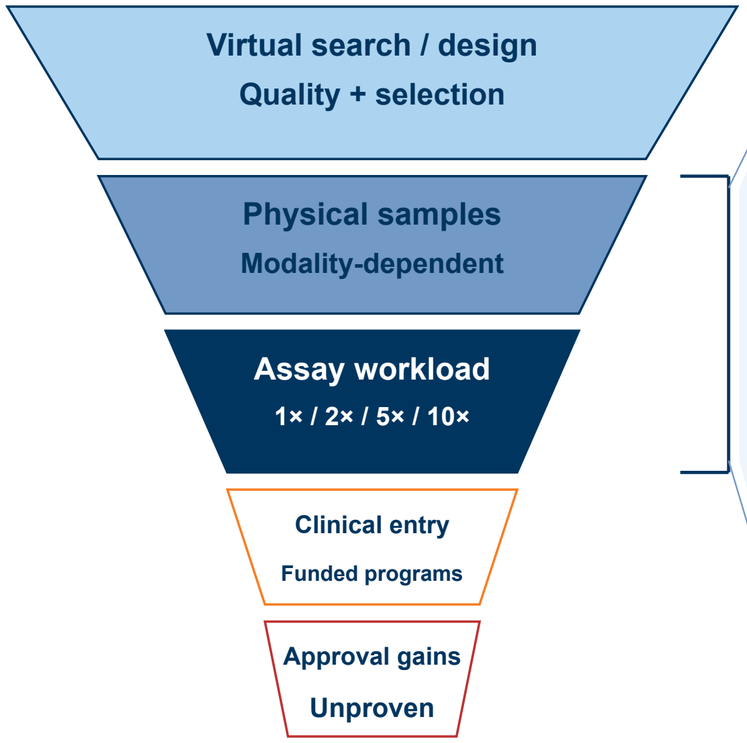
Sources: Lyu 2019; Sumitomo 2021; BIO 2011–2020; Wong 2000–2015. Workloads are analyst sensitivities, not empirical cohorts.

AI selection and biological evidence shape tool demand

Workflows vary by modality; paid demand depends on funded assays, depth, net inputs and prices.

MODALITY-AWARE DISCOVERY WORKFLOW

Schematic widths; workload multipliers are sensitivities



SMALL MOLECULES

Fewer syntheses per program are possible.

PROTEINS / ANTIBODIES

Pooled screens scale; individual profiling differs.

WET LAB = BUILD + TEST
No universal molecule-count funnel.
AI clinical advantage remains unproven.

The LST ecosystem behind the capacity gate

Issuer-cited subset estimates · different scopes / years · source conflicts disclosed below

Molecular construction

DNA / RNA synthesis, enzymes
Chemical building blocks

RMB 1.33bn

China • 2025 • DNA synthesis services [1]
Oligos + long fragments only

TWIST BIOSCIENCE · GenScript Make Research Easy

CHINA / HK LISTINGS	US LISTINGS / PARENTS
GenScript · 1548.HK	Twist · TWST
Vazyme · 688105.SH	Thermo Fisher · TMO
Tsingke · NEEQ:874813	

Protein & functional assays

Recombinant proteins, antibodies
Binding and activity assays

US$3.5bn

Global · 2025 · F&S RUO proteins [2]
Issuer-commissioned; estimate disputed

Acro BIOSYSTEMS

CHINA / HK LISTINGS	US LISTINGS / PARENTS
ACRO · 301080.SZ	Bio-Techne · TECH
Sino · 301047.SZ	Thermo Fisher · TMO

Cells & culture systems

Cell models, media, feeds
Bioreactors and cell expansion

Source pending

China • 2024 • bioprocess media [3]
Original filing unavailable; TAM withheld

Gibco / HyClone

CHINA / HK LISTINGS	US LISTINGS / PARENTS
OPM · 688293.SH	Gibco / Thermo · TMO
ACRO · 301080.SZ	Cytiva / HyClone · DHR

Labware & sample handling

Tips, plates, tubes, filtration
Automation-compatible labware

~RMB 23bn

China • 2024 • plastic labware [4]
Includes clinical and industrial uses

BIOFIL · BD

CHINA / HK LISTINGS	US LISTINGS / PARENTS
JET Biofil · 688026.SH	Corning · GLW
Changhong · 300151.SZ	BD · BDX

Measurement & analytics

Sequencing, LC / LC-MS, flow
Imaging and quality control

RMB 4.3bn

China • 2024 • sequencing tools [5]
Sequencing subset; 2024 policy baseline

MGI · BD flow → WAT†

CHINA / HK LISTINGS	US LISTINGS / PARENTS
MGI · 688114.SH	Illumina · ILMN
NanoMicro · 688690.SH	Agilent · A
	Waters · WAT†

Purification & scale-up

Analytical columns, process resins
Filtration and single-use systems

Source pending

China • 2024 • bioprocess resins [3]
Original filing unavailable; TAM withheld

NanoMicro · cytiva

CHINA / HK LISTINGS	US LISTINGS / PARENTS
NanoMicro · 688690.SH	Cytiva · DHR
Sepax · 688758.SH	Repligen · RGEN

Automation + data orchestration span every tool family — TECAN. SIX: TECN

† Ex-BD biosciences / flow moved to Waters on 9 Feb 2026; BD (BDX) here means specimen collection.
Protein TAM is issuer-commissioned / disputed. Media & resin TAMs withheld. Sources, scope and tickers: SOURCES.md.

01 The life science tools framework

FRAMEWORK

A product framework spanning academic research, biotech, pharma, CRDMOs and manufacturing.

Reagents

01	Recombinant proteins / antibodies
02	Enzymes / PCR and NGS reagents
03	DNA / RNA / chemical building blocks
04	Cell lines / culture media
05	Assay reagents for R&D and QC

Tests × dose per test × price

Consumables

01	Tips / microplates
02	Culture flasks / centrifuge tubes
03	Membranes / functional beads
04	Columns / chromatography resins
05	Gels / process components

Tests or batches × intensity × price

Instruments

01	Sequencers / liquid handlers
02	LC / LC-MS / mass spectrometers
03	Flow cytometers
04	High-content imaging systems
05	Cell-culture workstations

New installations + replacement + service

Use one product taxonomy; retain the scope of each market estimate.

Classify products separately from their use cases. Chromatography resins can be reused.

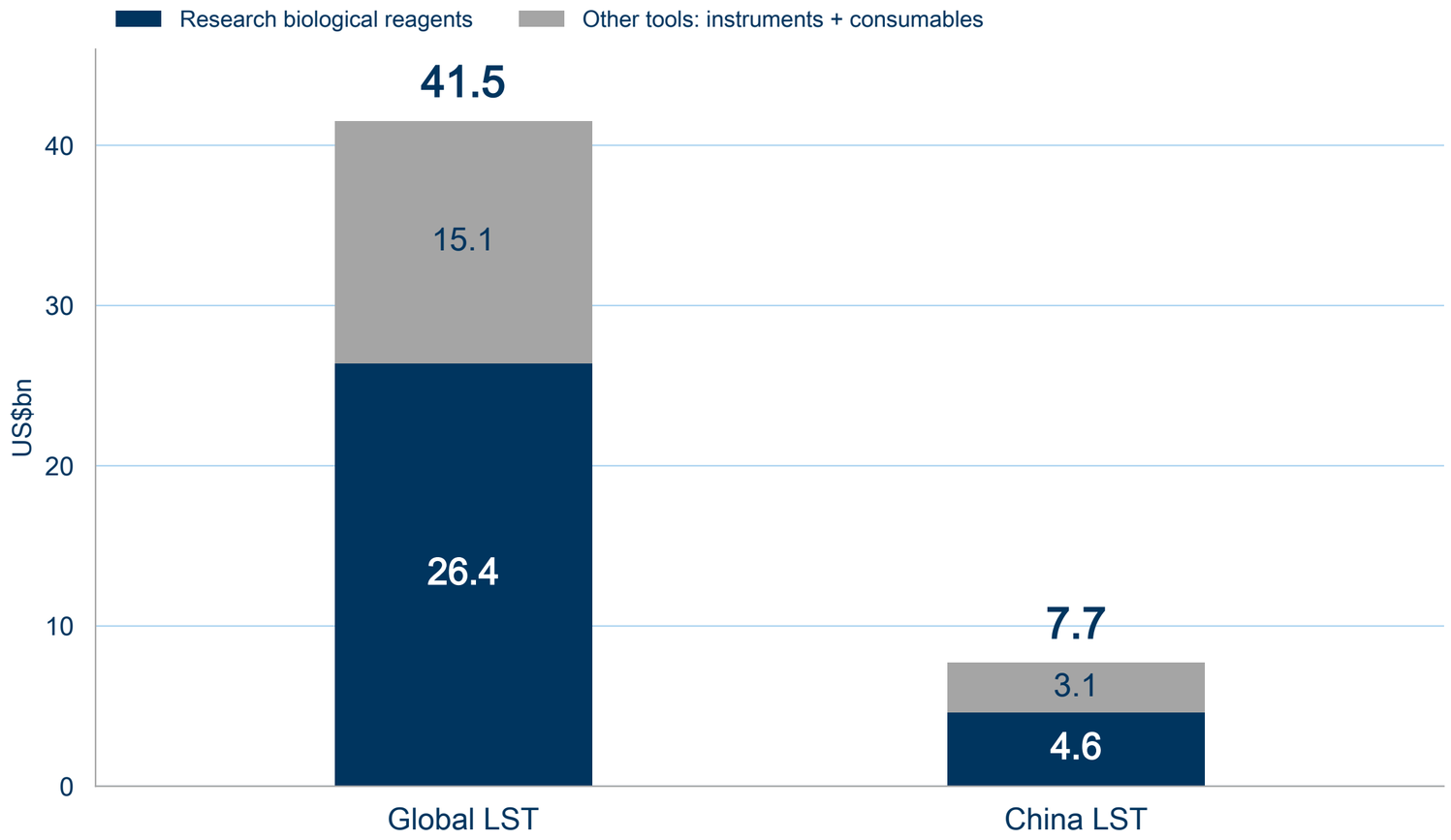
Source: supplied framework and prior product research. Existing TAMs do not automatically reconcile.

2026-10-12

02 2025 LST under one F&S definition

HISTORICAL

2025 F&S estimates commissioned for ACRO’s filing. A defined scope, not a broad-industry TAM.



China share within F&S scope

18.6%

7.7 / 41.5
Not vendor revenue exposure

Issuer-commissioned estimate.
Broader markets and vendor
China revenues use other scopes.

Other tools are the residual. Comparable instrument and consumable splits were not separately available.

Source: ACROBiosystems HKEX application, 30 Sep 2026, printed pp. 65 and 67; links in SOURCES.md.

03 Customer workflows across life science tools

USE MAP

A qualitative workflow map. Customer purchasing shares have not been established.

CUSTOMER	REAGENTS	CONSUMABLES	INSTRUMENTS
Academia / research hospitals	Targets, enzymes, cell media	Plates, tips, cultureware	Sequencing, imaging, LC-MS
Biotech / pharma R&D	Proteins, assay panels, expression reagents	Sensors, columns, filtration	BLI / SPR, automation, flow cytometry
CRO / CDMO	Discovery and process reagents	Labware, resins, scale-specific inputs	Assay, process and QC platforms
Biomanufacturing	Production media, feeds and enzymes	Resins, filters, single-use assemblies	Bioreactors, chromatography, QC instruments
IVD development	Antigens, antibodies, tool enzymes	Assay plates, magnetic beads	Research and QC instruments

An instrument supplier’s total China sales are not its China life-science sales.

Food, environmental and chemical-analysis uses excluded. Research instruments exclude clinical IVD analyzers.

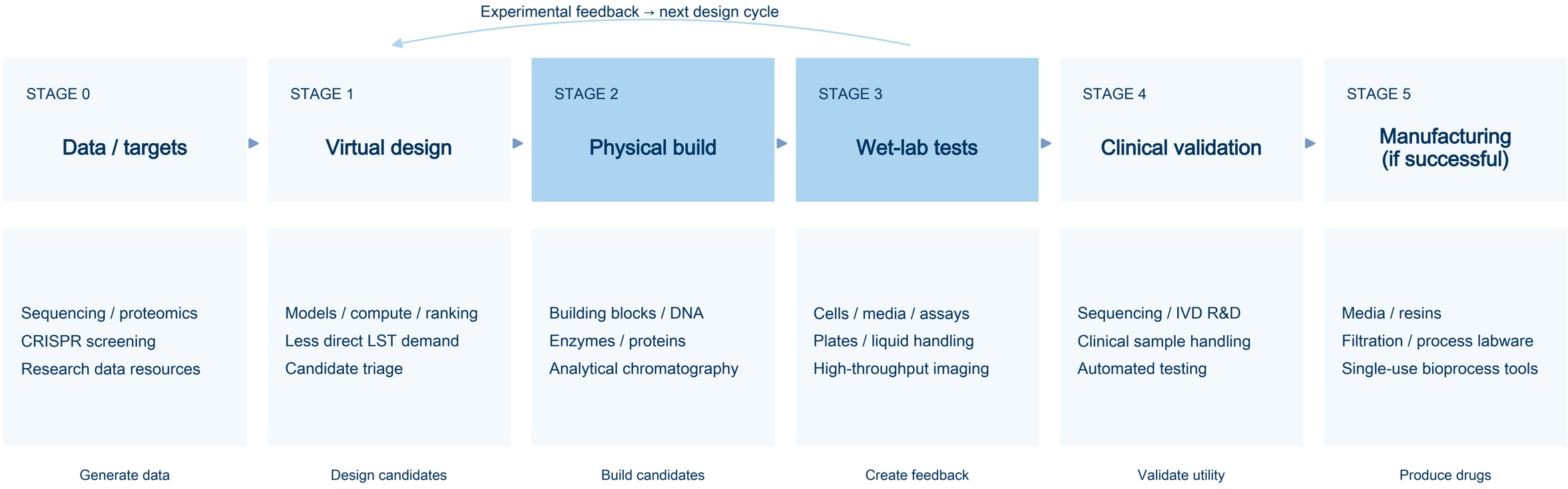
Source: representative supplier product uses; a role map, not an estimated customer revenue mix.

2026-10-13

04 AI discovery and its LST touchpoints

MECHANISM

AI candidates need physical construction and experimental feedback. Orders depend on funded projects.



The most direct potential uplift comes from building and testing candidates.

Stages 2–3 have more direct exposure. Clinical and manufacturing demand face attrition and time lags.

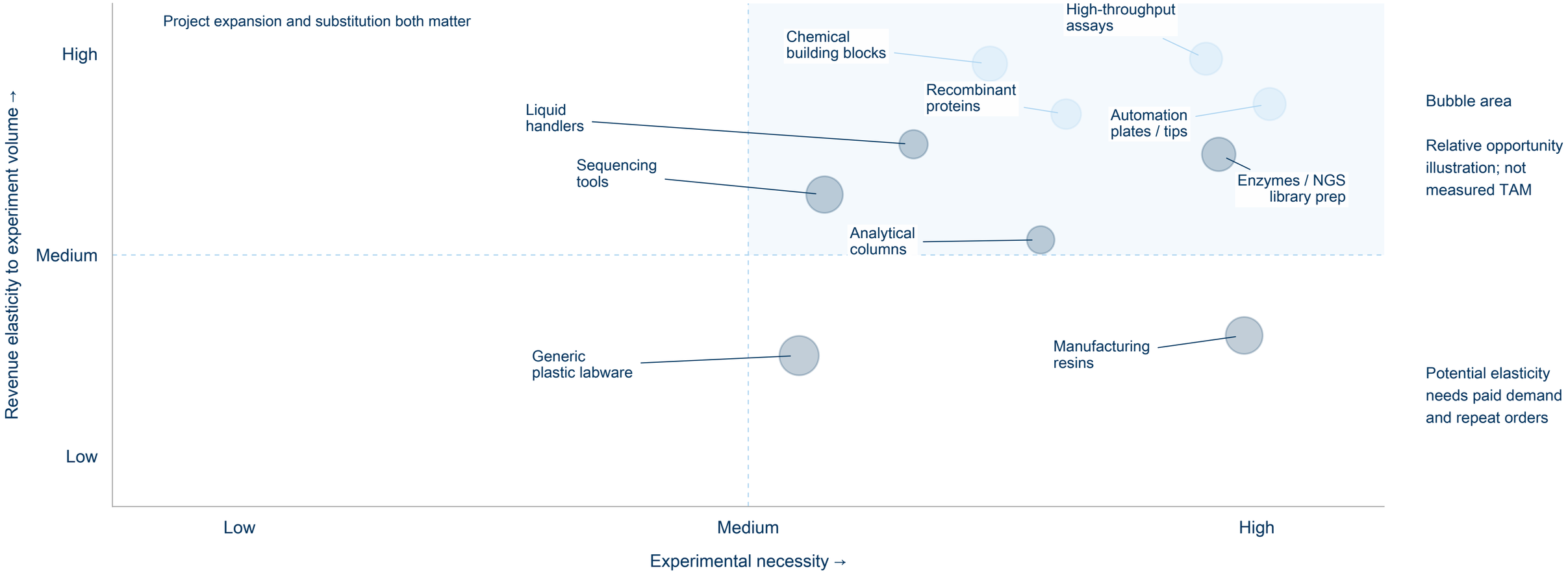
Source: supplied framework and product-use research. Arrows describe a mechanism, not realized orders.

2026-10-12

05 Which tools have more demand elasticity?

HYPOTHESIS

Positions and bubble areas are qualitative research judgments, not measured scores or additive TAMs.



Small-molecule synthesis can fall per project. Modality, assay depth, unit use and funded projects determine orders.

Source: supplied framework and analyst assumptions. Edit positions and relative areas in data.json.

2026-10-13

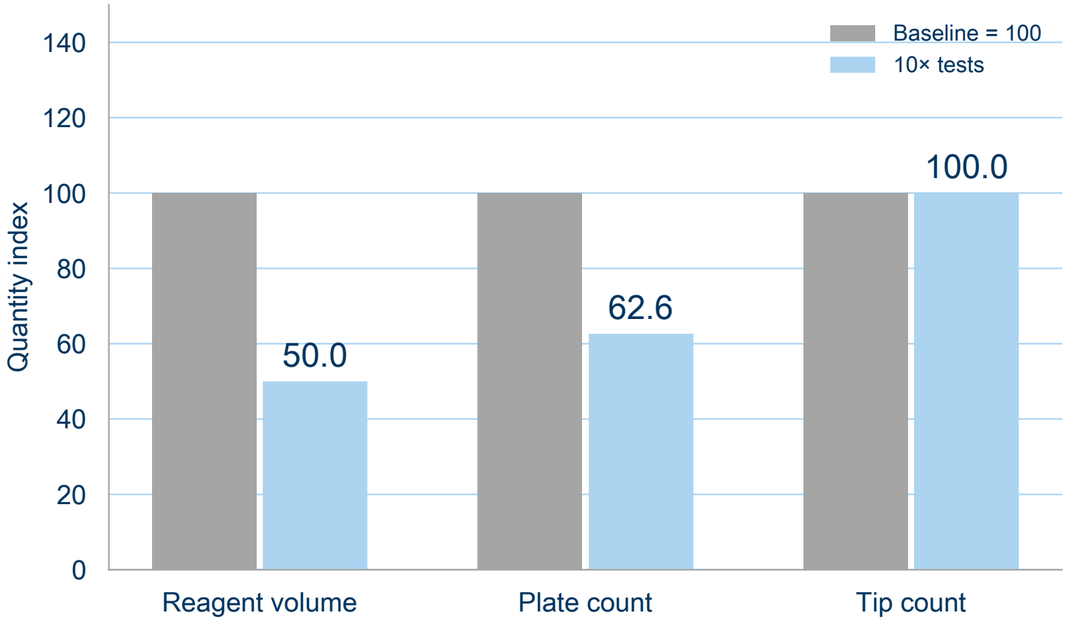
06 Quantities and prices in miniaturization

SENSITIVITY

10× tests: 10 L → 5 L reagent; 1,042 → 652 plates; 100k → 100k tips. Full usable plates.

Quantities × net unit prices determine spending

Physical inputs



Fewer plates can cost more

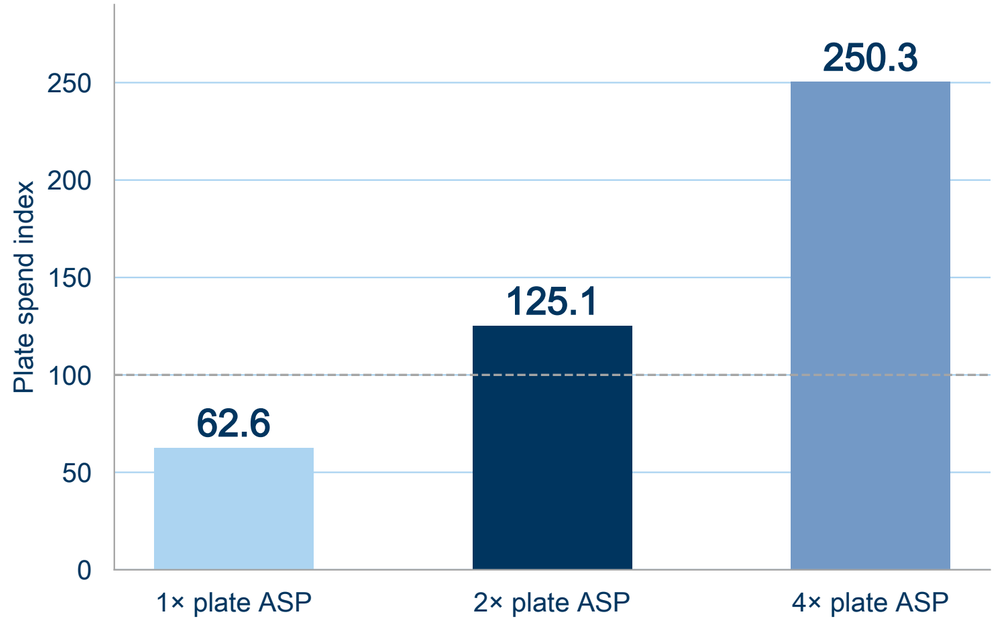


Plate spending exceeds baseline if the plate ASP ratio is above 1.60×.

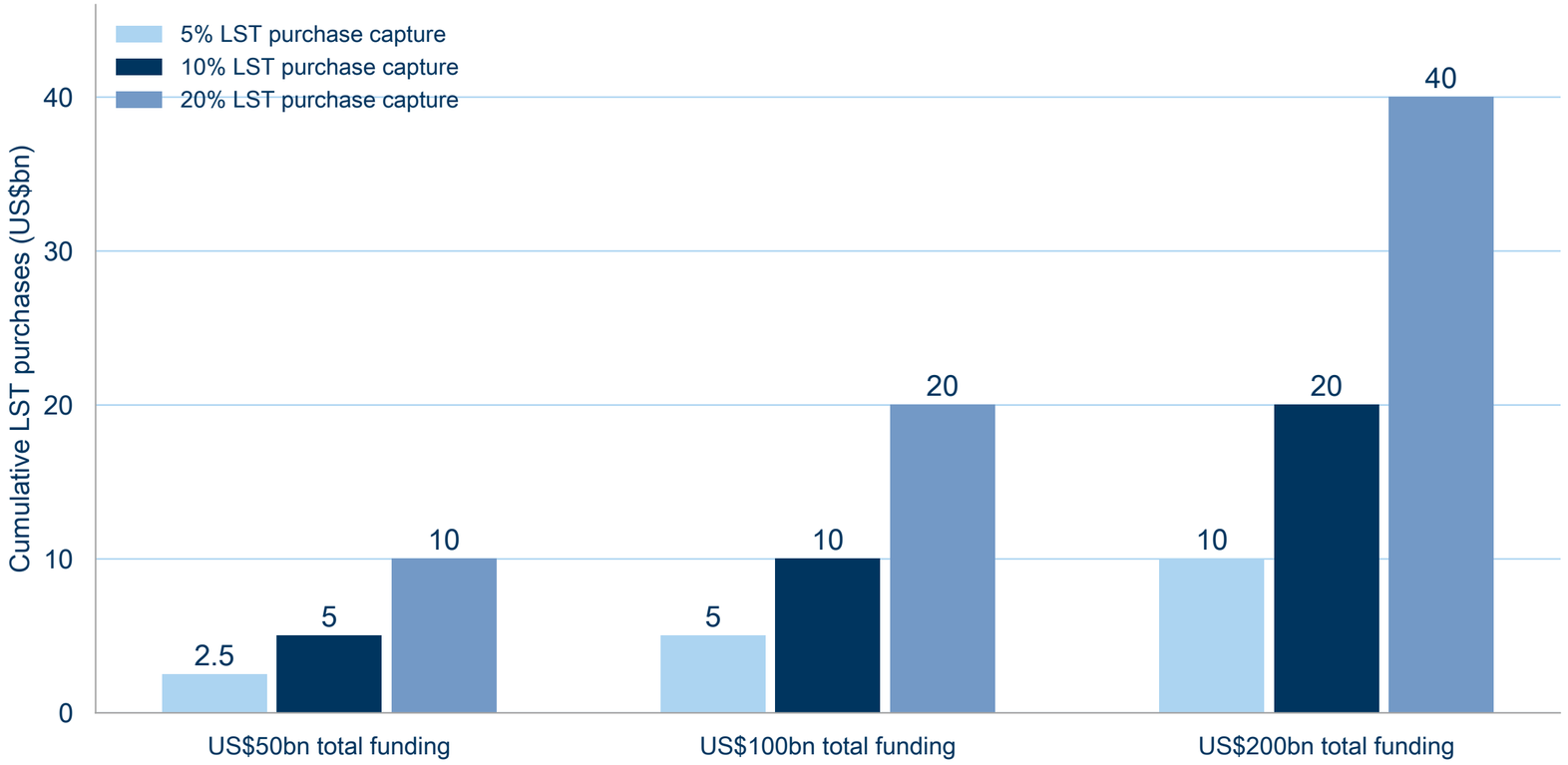
Price ratios are analyst inputs, not quoted ASPs. Dead volume, controls, usable wells and retesting can change quantities.

Source: transparent arithmetic from supplied test volumes and well counts. Units and sensitivities in SOURCES.md.

07 Funding-to-LST purchase sensitivity

SENSITIVITY

Hypothetical total funding pools. Only a fraction becomes purchases from LST suppliers.



Funding → tool purchases

R&D and production budgets exclude people, trials, software and unspent cash.

Capture includes tool purchases only. Count CRO/CDMO inputs once, separate from service fees.

Total capital raised is not an addressable tool market.

5% / 10% / 20% capture rates are model inputs. Cumulative scenarios, no fixed horizon; not observed funding or annual TAM.

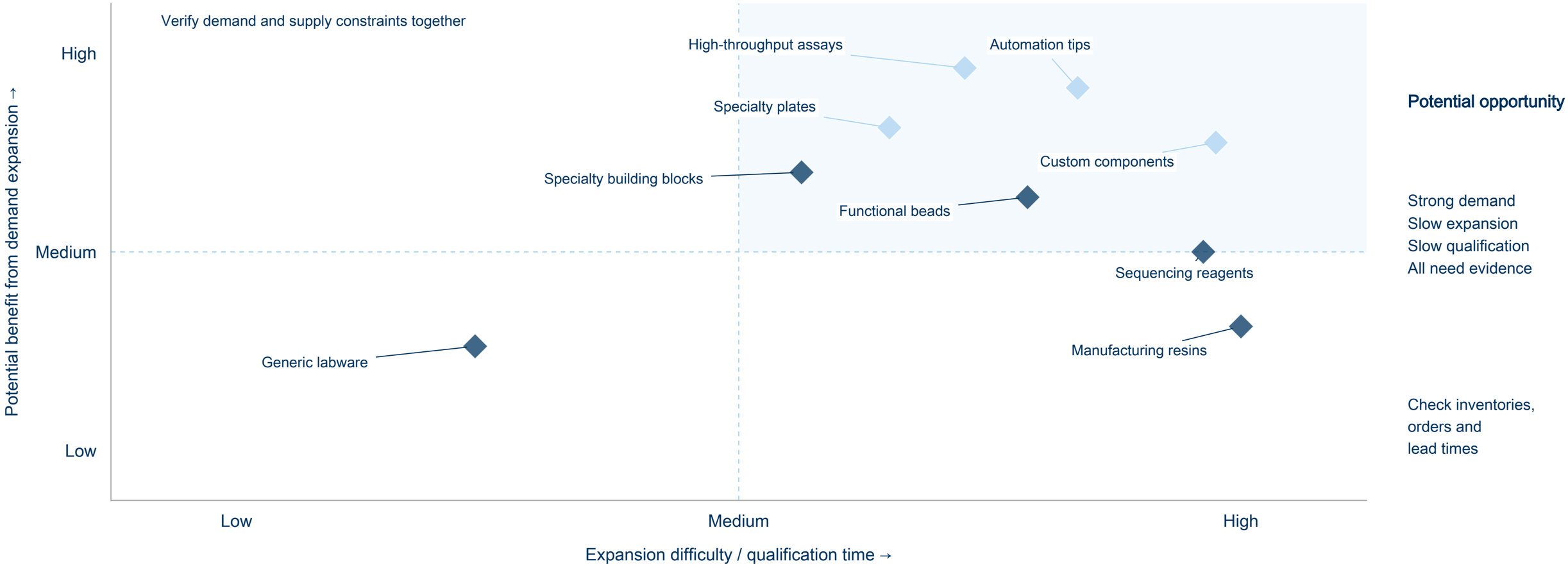
Source: analyst funding-to-purchase sensitivity. No empirical budget shares or equipment/reagent split is asserted.

2026-10-13

08 Potential supply constraints over 12–24 months

HYPOTHESIS

Relative expansion and qualification constraints over 12–24 months; current shortages are not established.



Track lead times, stockouts, qualification, yields and prices. Manufacturing resins have a longer demand lag.

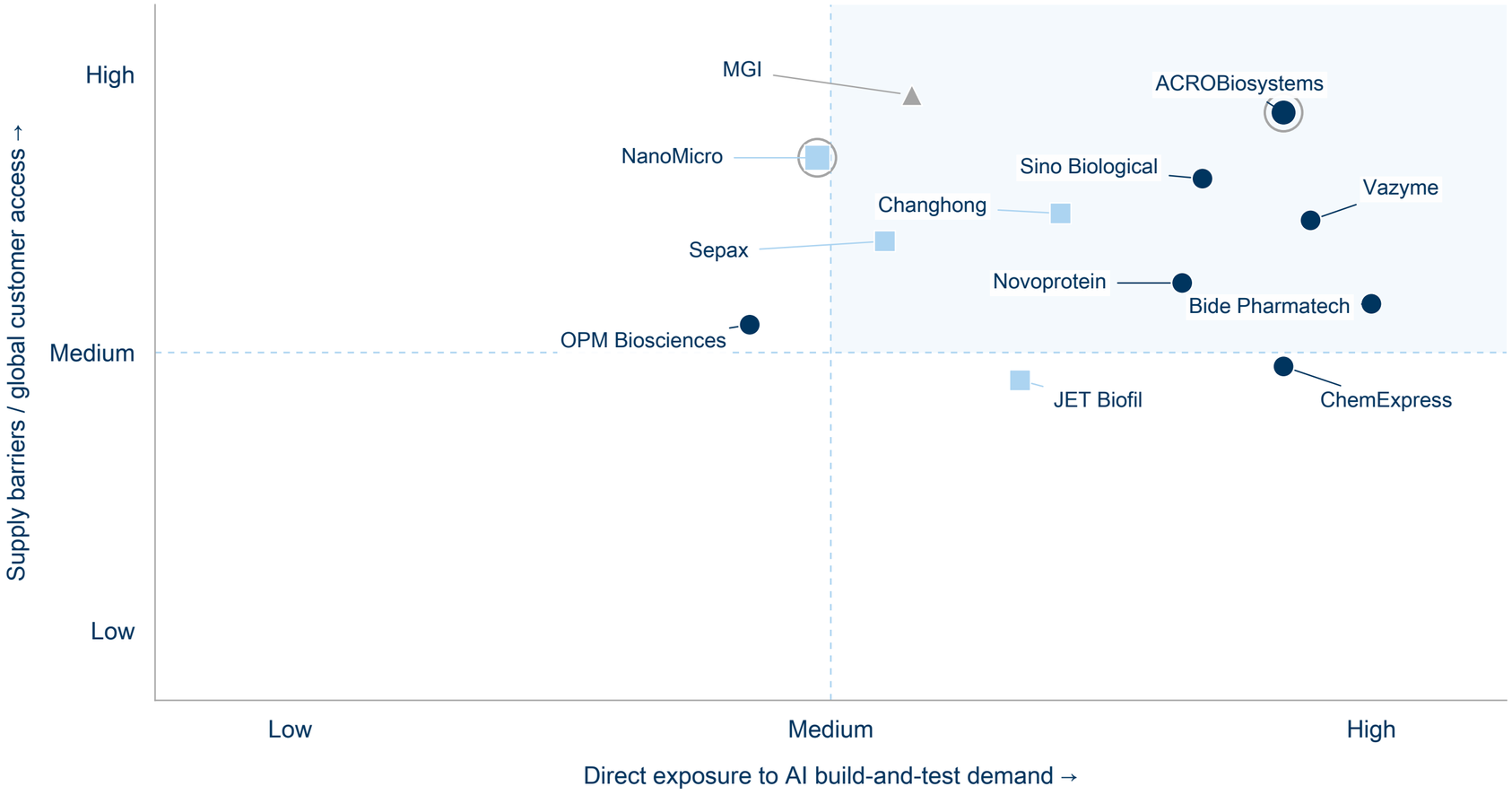
Source: supplied framework and analyst hypotheses. Technical barriers do not prove a capacity bottleneck.

2026-10-12

09 Mapping Chinese listed LST suppliers

HYPOTHESIS

Product exposure to AI build-and-test stages; positions do not measure AI revenue share or investment rank.



Reagents

Proteins, enzymes, blocks, media

Consumables

Labware, columns, resins

Instruments / platforms

Sequencing and automation

Two different demand pathways

ACRO: expression / functional assays
NanoMicro: purification + scale-up

Axes are qualitative judgments. Qualification, global customer access and paid orders still require evidence.

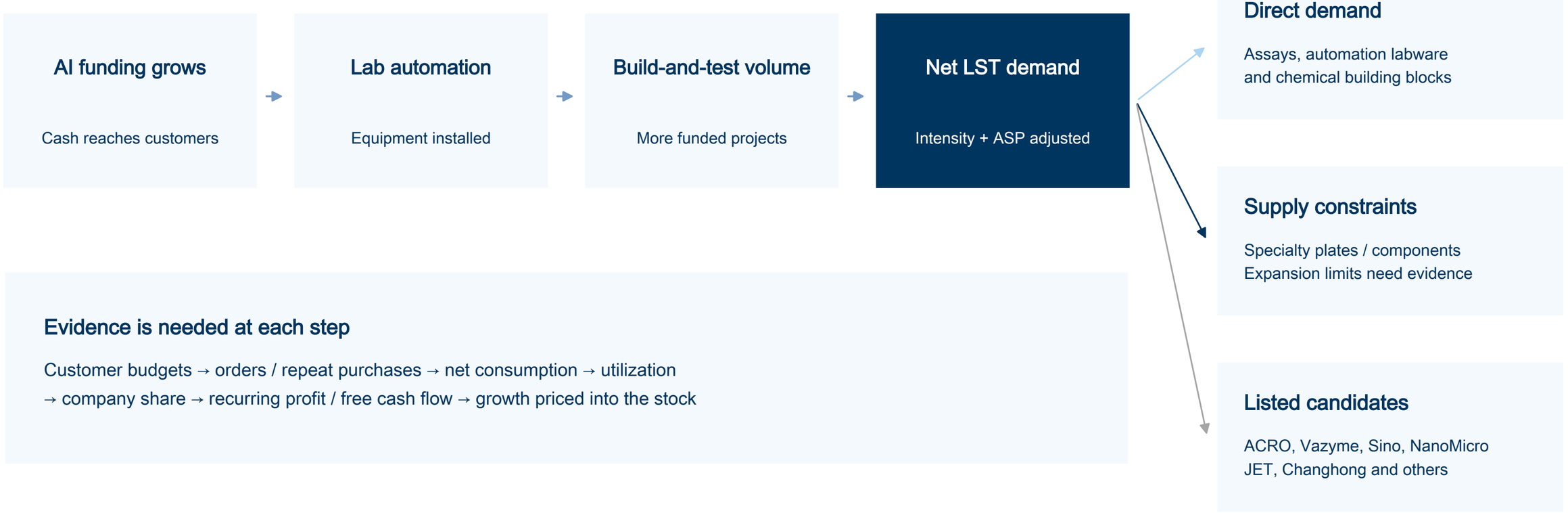
Sources: company filings and official product catalogs. Product rationale and assumptions are in data.json.

2026-10-12

10 From AI funding to LST shareholder returns

THESIS

Funding starts the chain. Projects, consumption, supply, market share and cash returns determine the outcome.



Evidence is needed at each step

Customer budgets → orders / repeat purchases → net consumption → utilization
→ company share → recurring profit / free cash flow → growth priced into the stock

Experimental relevance ∩ Evidenced supply constraints ∩ Global customer access

Seek experimental relevance, evidenced supply constraints and global access; test the valuation hurdle.

Source: supplied framework and prior company research. Capital announcements are not LST orders.

2026-10-12